# TRAN DAI NHAN

AI Engineer | Applied NLP, LLM Systems and Backend Services

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## EDUCATION

**FPT University** **Oct 2022 - Jan 2026**

*Bachelor of Engineering, Software Engineering. Major in Software Development* *GPA 3.3 / 4.0*

- Capstone: S.M.I.L.E, a multi clinic dental practice management platform. Owned appointment booking, payments and identity verification across a microservice backend.
- Coursework: distributed systems, database systems, software architecture, machine learning, computer vision, data structures and algorithms.
- Produced two accepted conference papers, founded an AI startup and led the university IT club alongside full time study.

## TECHNICAL SKILLS

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| **Languages** | Python, TypeScript, Java, C#, SQL, Bash |
| **AI and ML** | PyTorch, Transformers, vLLM, Unsloth, LLaMA-Factory, MS-SWIFT, LangGraph, LangChain, LangSmith, OpenAI API, NeMo Guardrails, Qdrant, Hugging Face |
| **AI Practice** | LLM fine tuning (SFT, GRPO), knowledge distillation, RAG and retrieval evaluation, agentic workflows, multimodal VQA, vision language pretraining, inference cost optimisation |
| **Backend** | FastAPI, NestJS, Spring Boot, .NET, REST API design, microservices, concurrency control, PostgreSQL, Redis, TypeORM |
| **Cloud and DevOps** | AWS (EC2, S3, IAM), Docker, Docker Compose, GitHub Actions CI/CD, Nginx, Cloudflare, Vercel, Linux, Oracle Cloud, Caddy, OpenTelemetry, PostHog |
| **Practices** | Agile and Scrum, code review, automated testing (Jest, pytest, Vitest) |

## EXPERIENCE

**Vingroup - AI Talent Training Program** **Sep 2026 - Present**

*AI Trainee, Team Lead of BRIDGE. LangGraph, OpenAI, NeMo Guardrails, FastAPI, Oracle Cloud* *Live demo*

- Led a four member team developing BRIDGE, a deployed research assistant for citation grounded literature review and code reproducibility assessment; responsible for system architecture and deployment.
- Designed a LangGraph agent pipeline with human approval checkpoints and resumable execution, combining retrieval from Semantic Scholar, OpenAlex and arXiv with criterion level LLM screening and full text verification.
- Built the production backend: a queue worker that resumes interrupted reviews from checkpoints, usage credits metered on actual LLM cost, and self hosted CI/CD that gates each commit on tests and rolls back failed releases.
- Benchmarked against Ai2's Asta Paper Finder on held out splits: recall of 46.7% versus 36.7% on RealScholarQuery and 19.5% versus 14.8% on ScholarQuest (F1 gains significant at $p < 0.001$), and parity on LitSearch at 76.2%.
- Evaluated the static reproducibility audit under the ReproRepo protocol on user reported issues from held out ICLR 2026 repositories, reaching SM@10 of 0.70 and EM@10 of 0.50 at under USD 0.03 per audit.
- Operated the service on Oracle Cloud with NeMo Guardrails, OpenTelemetry tracing, PostHog product analytics and LangSmith experiment tracking.

**AI VIET NAM** **Oct 2025 - Present**

*AI Researcher, Teaching Assistant and Research Team Lead* *Ho Chi Minh City*

- Supervise a 12 member research group in medical image analysis under the Research Mentorship Program 2026 (Dendrite Intelligence Lab), defining the research agenda, establishing selection criteria and reviewing experimental design and findings.
- Conduct empirical research in Vietnamese VQA and NLP, spanning dataset curation, model training, evaluation protocols and reporting of results.
- Develop course materials in NLP and multimodal AI, including lecture slides, lab notebooks and coding exercises, and guide learners in reproducing recent research.

**FPT Software HCM** **Jan 2025 - Jul 2025**

*Web Developer Intern, Backend* *Ho Chi Minh City*

- Built backend modules for a course management platform in Java and Spring Boot over PostgreSQL, adding Redis caching on frequently read data to cut response times.
- Automated delivery with a GitHub Actions CI/CD pipeline that builds Docker images and deploys on every merge, with the service exposed through Cloudflare, giving the team repeatable releases across Agile sprints.

## RESEARCH AND PUBLICATIONS

**Difficulty-Aware Adaptive Reasoning for Vietnamese VQA with GPT-OSS** **Accepted, AICI 2026**
*First author. Springer book chapter. PyTorch, Transformers, vLLM, Unsloth, Gemini 2.5* *Code*

- Designed a modular Vietnamese VQA framework covering dense image captioning, multi LLM inference across GPT-OSS, Qwen3 and DeepSeek, batch processing and automated evaluation.
- Built a difficulty aware router that scales inference compute per question across low, high and adaptive modes, cutting end to end inference from 3,317s to 815s, a 75 percent reduction against the high compute baseline.
- Reached BLEU@4 0.142, ROUGE-L 0.654 and METEOR 0.421 on ViVQA-X at a quarter of the baseline compute budget.

**Curating Multi-Mode CoT for Efficient Math Reasoning with GPT-OSS** **Accepted, ICISN 2026**
*Co-author. LLaMA-Factory, MS-SWIFT, GPT-OSS, Llama 3.2* *Code*

- Built the distillation pipeline curating low, medium and high budget reasoning traces from a GPT-OSS teacher for a Llama 3.2 3B student.
- Added final answer verification and length based filtering to strip noisy and over long chain of thought supervision, prioritising sample quality over volume.
- Ran SFT and GRPO workflows reaching GSM8K 0-shot 0.8006 and MATH500 0-shot 0.4760.

## SELECTED PROJECTS

**S.M.I.L.E, Dental Practice Management Platform** | Capstone **2026**
*Booking, Payment and Identity Owner. NestJS, Bun, TypeORM, PostgreSQL, Redis, LangGraph, Qdrant, Python* *Code*

- Built appointment booking APIs for three booking modes (by clinic, specialty or doctor) with database level concurrency control on doctor time slots, so simultaneous requests for one slot yield exactly one booking.
- Owned the payment service in a platform of four NestJS services and five PostgreSQL databases: VNPay QR and redirect flows, signed callbacks with replay protection, refunds, and Redis backed idempotency keys so client retries never double charge.
- Delivered a LangGraph booking assistant with Qdrant retrieval over clinic policies, and a KYC service (YOLOv11 and VietOCR) that extracts seven identity fields from both sides of a Vietnamese citizen ID card then cross check.

**Vietnamese Vision Language Model Pretraining from Scratch** **2026**
*Team project. SigLIP2, MLP projector, Llama 3.2 1B Instruct, PyTorch, Accelerate, Hugging Face* *Code*

- Implemented a two stage LLaVA style recipe: projector only alignment between a frozen SigLIP2 encoder and Llama decoder, then Vietnamese visual instruction tuning on scraped and OpenAI Batch generated data.
- Trained with Accelerate in a distributed, resumable setup, and evaluated the model on benchmarks and in demos for image captioning and visual chat.

**Vietnamese GPT-2 Pretraining from Scratch** **2026**
*Team project. PyTorch, Transformers, FastAPI, Docker, Next.js* *Code*

- Trained a Vietnamese subword tokenizer from scratch to reduce token fragmentation, and curated a deduplicated news and Wikipedia corpus of roughly 2.64B tokens for pretraining and continued pretraining.
- Conducted a controlled study of how continued pretraining on five word quatrain poetry shifts generation style, and served the model via a Dockerised FastAPI backend and a Next.js chat interface.

**PawCal, AI Veterinary Automation Platform** **2025 - Present**
*Founder and Technical Lead. Next.js, Supabase, n8n, Zalo OA, Vercel, Docker*

- Built a chatbot agent on Zalo OA handling appointment scheduling, automated reminders and customer enquiries for veterinary clinics.
- Secured roughly USD 2,000 in early support and validated the product with beta clinics.

## LEADERSHIP, AWARDS AND CERTIFICATIONS

**President, FCODER, FPT University Information Technology Club** **Oct 2023 - Oct 2025**

- Directed club operations, coding competitions and technical training; ranked 1st among academic clubs at FPT University in 3 of 4 terms.
- Proposed and led project initiatives presented to the leadership board of FPT University's academic clubs union.

**Research Leader, FPT Research Festival** **2025**

- Led a student research team through problem framing, experiment planning and paper writing, delivering an image labelling pipeline for breast cancer data and an optimised clinical prediction workflow across 2 accepted papers.

**Awards** | First Prize (team), Coca-Cola environmental protection solution challenge. Top 2, FPT Can Tho AI Student Olympiad 2026. Top 10, FPT Hackathon (AI support platform for children with autism). Provincial Excellent Student Award, Chemistry.
**Certifications and Languages** | Software Development Lifecycle (Coursera). MOS Word 950, Excel 975. Vietnamese native, English professional working proficiency.